Namespace ASE_Assignment

Classes

AppCanvas

Initialises drawing surface and provides methods for drawing and also manages the drawing surface.

AppCommandFactory

Factory class for creating instances of ASE_Assignment and BOOSE commands. BOOSE.CommandFactory

Circle

Command to draw a circle on the canvas.

Form1

Main form of the application. Handles user input and initialising drawing canvas.

MoveTo

Command which moves the pen to specified (x,y) coordinates.

PenColour

Command which sets the pen colour to draw with using RGB values.

Rect

Command to draw a rectangle on the canvas.

Class AppCanvas

Namespace: ASE_Assignment

Assembly: ASE_Assignment.dll

Initialises drawing surface and provides methods for drawing and also manages the drawing surface.

```
public class AppCanvas : Canvas, ICanvas
```

Inheritance

object ← Canvas ← AppCanvas

Implements

ICanvas

Inherited Members

Canvas.background_colour , Canvas.Tri(int, int) , Canvas.Rectangle(int, int, bool) , Canvas.Reset() , Canvas.WriteText(string) , Canvas.Xpos , Canvas.Ypos , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AppCanvas(int, int)

Initialises an instance of the AppCanvas class using the width and height provided.

```
public AppCanvas(int width, int height)
```

Parameters

`width` int

Width of the drawing surface

`height` int

Height of the drawing surface

Properties

PenColour

Sets the pen colour used when drawing on the canvas.

```
public override object PenColour { get; set; }
```

Property Value

object

Methods

CheckWithinBounds(int, int)

Checks if parameters given are within drawing surface bounds and throws BOOSE.CommandException if not.

```
public void CheckWithinBounds(int x, int y)
```

Parameters

x int

X position to be checked if within bounds.

y int

Y position to be checked if within bounds.

Exceptions

CommandException

Thrown if X or Y provided are out of bounds.

Circle(int, bool)

Draws a circle from the current position with the specified radius, filled or unfilled.

```
public override void Circle(int radius, bool filled)
```

Parameters

`radius` int

Radius of the circle drawn

`filled` bool

Bool representing if the circle should be filled

Clear()

Clears the canvas by filling it with the current background colour.

```
public override void Clear()
```

DrawTo(int, int)

Draws a line from the current position to the specified (x,y) coordinates and sets the current position to the (x,y) coordinates.

```
public override void DrawTo(int x, int y)
```

Parameters

`x` int

X coordinate to draw to

`y` int

Y coordinate to draw to

MoveTo(int, int)

Moves pen position to given (x,y) coordinate.

```
public override void MoveTo(int x, int y)
```

Parameters

`x` int

X position to move to.

`y` int

Y position to move to.

Rect(int, int, bool)

Draws a rectangle with the specified width and height, filled or unfilled.

```
public override void Rect(int width, int height, bool filled)
```

Parameters

`width` int

Width of the rectangle

`height` int

Height of the rectangle

`filled` bool

Bool representing if the rectangle should be filled

Set(int, int)

Initialises then sets the default conditions of the canvas, such as background colour and pen colour.

```
public override void Set(int width, int height)
```

Parameters

`width` int

Width of the drawing surface

`height` int

Height of the drawing surface

SetColour(int, int, int)

Sets the pen colour using RGB values.

```
public override void SetColour(int red, int green, int blue)
```

Parameters

`red` int

Red value of pen colour

`green` int

Green value of pen colour

`blue` int

Blue value of pen colour

getBitmap()

Used to retrieve the bitmap of the AppCanvas drawing surface

```
public override object getBitmap()
```

Returns

object

Object of the drawing surface

Class AppCommandFactory

Namespace: ASE_Assignment

Assembly: ASE_Assignment.dll

Factory class for creating instances of ASE_Assignment and BOOSE commands. BOOSE.CommandFactory

```
public class AppCommandFactory : CommandFactory, ICommandFactory
```

Inheritance

object ← CommandFactory ← AppCommandFactory

Implements

ICommandFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

MakeCommand(string)

Creates an instance of specified command based on commandType provided.

```
public override ICommand MakeCommand(string commandType)
```

Parameters

`commandType` string

String correlating to class constructor to be initialised. If commandType is not recognised BOOSE.CommandFactory is called to determine if the command exists within BOOSE library.

Returns

ICommand

Returns ICommand object corresponding to the commandType

See Also

ICommand

Class Circle

Namespace: ASE_Assignment

Assembly: ASE_Assignment.dll

Command to draw a circle on the canvas.

```
public class Circle : CommandOneParameter, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← Circle

Implements

ICommand

Inherited Members

CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Circle()

Initialises new instance of Circle with no parameters.

```
public Circle()
```

Circle(Canvas, int)

Initialises new instance of Circle with a specified canvas and radius.

```
public Circle(Canvas c, int radius)
```

Parameters

`c` Canvas

Canvas that the circle is to be drawn on.

`radius` int

The radius of the circle that will be drawn.

Methods

CheckParameters(string[])

Checks if Circle command has been called with the correct amount of parameters.

```
public override void CheckParameters(string[] parameterList)
```

Parameters

`parameterList` string[]

String array to be checked to ensure the correct number of parameters has been passed.

Exceptions

CommandException

Thrown if the number of parameters is not 1.

Execute()

Calls BOOSE.Command.Execute(), then draws a circle with specified radius.

```
public override void Execute()
```

See Also

CommandOneParameter

Class Form1

Namespace: ASE_Assignment

Assembly: ASE_Assignment.dll

Main form of the application. Handles user input and initialising drawing canvas.

```
public class Form1 : Form, IDropTarget, ISynchronizeInvoke, IWin32Window,
IBindableComponent, IComponent, IDisposable, IContainerControl
```

Inheritance

object ← MarshalByRefObject ← Component ← Control ← ScrollableControl ← ContainerControl ← Form ← Form1

Implements

IDropTarget, ISynchronizeInvoke, IWin32Window, IBindableComponent, IComponent, IDisposable, IContainerControl

Inherited Members

Form.SetVisibleCore(bool) , Form.Activate() , Form.ActivateMdiChild(Form) , Form.AddOwnedForm(Form) , Form.AdjustFormScrollbars(bool) , Form.Close() , Form.CreateAccessibilityInstance() , Form.CreateControlsInstance() , Form.CreateHandle() , Form.DefWndProc(ref Message) , Form.ProcessMnemonic(char) , Form.CenterToParent() , Form.CenterToScreen() , Form.LayoutMdi(MdiLayout) , Form.OnActivated(EventArgs) , Form.OnBackgroundImageChanged(EventArgs) , Form.OnBackgroundImageLayoutChanged(EventArgs) , Form.OnClosing(CancelEventArgs) , Form.OnClosed(EventArgs) , Form.OnFormClosing(FormClosingEventArgs) , Form.OnFormClosed(FormClosedEventArgs) , Form.OnCreateControl() , Form.OnDeactivate(EventArgs) , Form.OnEnabledChanged(EventArgs) , Form.OnEnter(EventArgs) , Form.OnFontChanged(EventArgs) , Form.OnGotFocus(EventArgs) , Form.OnHandleCreated(EventArgs) , Form.OnHandleDestroyed(EventArgs) , Form.OnHelpButtonClicked(CancelEventArgs) , Form.OnLayout(LayoutEventArgs) , Form.OnLoad(EventArgs) , Form.OnMaximizedBoundsChanged(EventArgs) , Form.OnMaximumSizeChanged(EventArgs) , Form.OnMinimumSizeChanged(EventArgs) , Form.OnInputLanguageChanged(InputLanguageChangedEventArgs) , Form.OnInputLanguageChanging(InputLanguageChangingEventArgs) , Form.OnVisibleChanged(EventArgs) , Form.OnMdiChildActivate(EventArgs) , Form.OnMenuStart(EventArgs) , Form.OnMenuComplete(EventArgs) , Form.OnPaint(PaintEventArgs) , Form.OnResize(EventArgs) , Form.OnDpiChanged(DpiChangedEventArgs) , Form.OnGetDpiScaledSize(int, int, ref Size) ,

Form.OnRightToLeftLayoutChanged(EventArgs) , Form.OnShown(EventArgs) ,
Form.OnTextChanged(EventArgs) , Form.ProcessCmdKey(ref Message, Keys) ,
Form.ProcessDialogKey(Keys) , Form.ProcessDialogChar(char) ,
Form.ProcessKeyPreview(ref Message) , Form.ProcessTabKey(bool) ,
Form.RemoveOwnedForm(Form) , Form.Select(bool, bool) ,
Form.ScaleMinMaxSize(float, float, bool) ,
Form.GetScaledBounds(Rectangle, SizeF, BoundsSpecified) ,
Form.ScaleControl(SizeF, BoundsSpecified) , Form.SetBoundsCore(int, int, int, int, BoundsSpecified) ,
Form.SetClientSizeCore(int, int) , Form.SetDesktopBounds(int, int, int, int) ,
Form.SetDesktopLocation(int, int) , Form.Show(IWin32Window) , Form.ShowDialog() ,
Form.ShowDialog(IWin32Window) , Form.ToString() , Form.UpdateDefaultButton() ,
Form.OnResizeBegin(EventArgs) , Form.OnResizeEnd(EventArgs) ,
Form.OnStyleChanged(EventArgs) , Form.ValidateChildren() ,
Form.ValidateChildren(ValidationConstraints) , Form.WndProc(ref Message) , Form.AcceptButton ,
Form.ActiveForm , Form.ActiveMdiChild , Form.AllowTransparency , Form.AutoScroll ,
Form.AutoSize , Form.AutoSizeMode , Form.AutoValidate , Form.BackColor ,
Form.FormBorderStyle , Form.CancelButton , Form.ClientSize , Form.ControlBox ,
Form.CreateParams , Form.DefaultImeMode , Form.DefaultSize , Form.DesktopBounds ,
Form.DesktopLocation , Form.DialogResult , Form.HelpButton , Form.Icon , Form.IsMdiChild ,
Form.IsMdiContainer , Form.IsRestrictedWindow , Form.KeyPreview , Form.Location ,
Form.MaximizedBounds , Form.MaximumSize , Form.MainMenuStrip , Form.MinimumSize ,
Form.MaximizeBox , Form.MdiChildren , Form.MdiChildrenMinimizedAnchorBottom ,
Form.MdiParent , Form.MinimizeBox , Form.Modal , Form.Opacity , Form.OwnedForms ,
Form.Owner , Form.RestoreBounds , Form.RightToLeftLayout , Form.ShowInTaskbar ,
Form.ShowIcon , Form.ShowWithoutActivation , Form.Size , Form.SizeGripStyle ,
Form.StartPosition , Form.Text , Form.TopLevel , Form.TopMost , Form.TransparencyKey ,
Form.WindowState , Form.AutoSizeChanged , Form.AutoValidateChanged ,
Form.HelpButtonClicked , Form.MaximizedBoundsChanged , Form.MaximumSizeChanged ,
Form.MinimumSizeChanged , Form.Activated , Form.Deactivate , Form.FormClosing ,
Form.FormClosed , Form.Load , Form.MdiChildActivate , Form.MenuComplete ,
Form.MenuStart , Form.InputLanguageChanged , Form.InputLanguageChanging ,
Form.RightToLeftLayoutChanged , Form.Shown , Form.DpiChanged , Form.ResizeBegin ,
Form.ResizeEnd , ContainerControl.OnAutoValidateChanged(EventArgs) ,
ContainerControl.OnMove(EventArgs) , ContainerControl.OnParentChanged(EventArgs) ,
ContainerControl.PerformAutoScale() , ContainerControl.RescaleConstantsForDpi(int, int) ,
ContainerControl.Validate() , ContainerControl.Validate(bool) ,
ContainerControl.AutoScaleDimensions , ContainerControl.AutoScaleFactor ,
ContainerControl.AutoScaleMode , ContainerControl.BindingContext ,
ContainerControl.CanEnableIme , ContainerControl.ActiveControl ,
ContainerControl.CurrentAutoScaleDimensions , ContainerControl.ParentForm ,

ScrollableControl.ScrollStateAutoScrolling , ScrollableControl.ScrollStateHScrollVisible , ScrollableControl.ScrollStateVScrollVisible , ScrollableControl.ScrollStateUserHasScrolled , ScrollableControl.ScrollStateFullDrag , ScrollableControl.GetScrollState(int) , ScrollableControl.OnMouseWheel(MouseEventArgs) , ScrollableControl.OnRightToLeftChanged(EventArgs) , ScrollableControl.OnPaintBackground(PaintEventArgs) , ScrollableControl.OnPaddingChanged(EventArgs) , ScrollableControl.SetDisplayRectLocation(int, int) , ScrollableControl.ScrollControlIntoView(Control) , ScrollableControl.ScrollToControl(Control) , ScrollableControl.OnScroll(ScrollEventArgs) , ScrollableControl.SetAutoScrollMargin(int, int) , ScrollableControl.SetScrollState(int, bool) , ScrollableControl.AutoScrollMargin , ScrollableControl.AutoScrollPosition , ScrollableControl.AutoScrollMinSize , ScrollableControl.DisplayRectangle , ScrollableControl.HScroll , ScrollableControl.HorizontalScroll , ScrollableControl.VScroll , ScrollableControl.VerticalScroll , ScrollableControl.Scroll , Control.GetAccessibilityObjectById(int) , Control.SetAutoSizeMode(AutoSizeMode) , Control.GetAutoSizeMode() , Control.GetPreferredSize(Size) , Control.AccessibilityNotifyClients(AccessibleEvents, int) , Control.AccessibilityNotifyClients(AccessibleEvents, int, int) , Control.BeginInvoke(Delegate) , Control.BeginInvoke(Action) , Control.BeginInvoke(Delegate, params object[]) , Control.BringToFront() , Control.Contains(Control) , Control.CreateGraphics() , Control.CreateControl() , Control.DestroyHandle() , Control.DoDragDrop(object, DragDropEffects) , Control.DoDragDrop(object, DragDropEffects, Bitmap, Point, bool) , Control.DrawToBitmap(Bitmap, Rectangle) , Control.EndInvoke(IAsyncResult) , Control.FindForm() , Control.GetTopLevel() , Control.RaiseKeyEvent(object, KeyEventArgs) , Control.RaiseMouseEvent(object, MouseEventArgs) , Control.Focus() , Control.FromChildHandle(nint) , Control.FromHandle(nint) , Control.GetChildAtPoint(Point, GetChildAtPointSkip) , Control.GetChildAtPoint(Point) , Control.GetContainerControl() , Control.GetNextControl(Control, bool) , Control.GetStyle(ControlStyles) , Control.Hide() , Control.InitLayout() , Control.Invalidate(Region) , Control.Invalidate(Region, bool) , Control.Invalidate() , Control.Invalidate(bool) , Control.Invalidate(Rectangle) , Control.Invalidate(Rectangle, bool) , Control.Invoke(Action) , Control.Invoke(Delegate) , Control.Invoke(Delegate, params object[]) , Control.Invoke<T>(Func<T>) , Control.InvokePaint(Control, PaintEventArgs) , Control.InvokePaintBackground(Control, PaintEventArgs) , Control.IsKeyLocked(Keys) , Control.IsInputChar(char) , Control.IsInputKey(Keys) , Control.IsMnemonic(char, string) , Control.LogicalToDeviceUnits(int) , Control.LogicalToDeviceUnits(Size) , Control.ScaleBitmapLogicalToDevice(ref Bitmap) , Control.NotifyInvalidate(Rectangle) , Control.InvokeOnClick(Control, EventArgs) , Control.OnAutoSizeChanged(EventArgs) , Control.OnBackColorChanged(EventArgs) , Control.OnBindingContextChanged(EventArgs) , Control.OnCausesValidationChanged(EventArgs) , Control.OnContextMenuStripChanged(EventArgs) , Control.OnCursorChanged(EventArgs) , Control.OnDataContextChanged(EventArgs) ,

Control.OnDockChanged(EventArgs) , Control.OnForeColorChanged(EventArgs) ,
Control.OnNotifyMessage(Message) , Control.OnParentBackColorChanged(EventArgs) ,
Control.OnParentBackgroundImageChanged(EventArgs) ,
Control.OnParentBindingContextChanged(EventArgs) , Control.OnParentCursorChanged(EventArgs) ,
Control.OnParentDataContextChanged(EventArgs) , Control.OnParentEnabledChanged(EventArgs) ,
Control.OnParentFontChanged(EventArgs) , Control.OnParentForeColorChanged(EventArgs) ,
Control.OnParentRightToLeftChanged(EventArgs) , Control.OnParentVisibleChanged(EventArgs) ,
Control.OnPrint(PaintEventArgs) , Control.OnTabIndexChanged(EventArgs) ,
Control.OnTabStopChanged(EventArgs) , Control.OnClick(EventArgs) ,
Control.OnClientSizeChanged(EventArgs) , Control.OnControlAdded(ControlEventArgs) ,
Control.OnControlRemoved(ControlEventArgs) , Control.OnLocationChanged(EventArgs) ,
Control.OnDoubleClick(EventArgs) , Control.OnDragEnter(DragEventArgs) ,
Control.OnDragOver(DragEventArgs) , Control.OnDragLeave(EventArgs) ,
Control.OnDragDrop(DragEventArgs) , Control.OnGiveFeedback(GiveFeedbackEventArgs) ,
Control.InvokeGotFocus(Control, EventArgs) , Control.OnHelpRequested(HelpEventArgs) ,
Control.OnInvalidated(InvalidateEventArgs) , Control.OnKeyDown(KeyEventArgs) ,
Control.OnKeyPress(KeyPressEventArgs) , Control.OnKeyUp(KeyEventArgs) ,
Control.OnLeave(EventArgs) , Control.InvokeLostFocus(Control, EventArgs) ,
Control.OnLostFocus(EventArgs) , Control.OnMarginChanged(EventArgs) ,
Control.OnMouseDoubleClick(MouseEventArgs) , Control.OnMouseClick(MouseEventArgs) ,
Control.OnMouseCaptureChanged(EventArgs) , Control.OnMouseDown(MouseEventArgs) ,
Control.OnMouseEnter(EventArgs) , Control.OnMouseLeave(EventArgs) ,
Control.OnDpiChangedBeforeParent(EventArgs) , Control.OnDpiChangedAfterParent(EventArgs) ,
Control.OnMouseHover(EventArgs) , Control.OnMouseMove(MouseEventArgs) ,
Control.OnMouseUp(MouseEventArgs) ,
Control.OnQueryContinueDrag(QueryContinueDragEventArgs) ,
Control.OnRegionChanged(EventArgs) , Control.OnPreviewKeyDown(PreviewKeyDownEventArgs) ,
Control.OnSizeChanged(EventArgs) , Control.OnChangeUICues(UICuesEventArgs) ,
Control.OnSystemColorsChanged(EventArgs) , Control.OnValidating(CancelEventArgs) ,
Control.OnValidated(EventArgs) , Control.PerformLayout() , Control.PerformLayout(Control, string) ,
Control.PointToClient(Point) , Control.PointToScreen(Point) ,
Control.PreProcessMessage(ref Message) , Control.PreProcessControlMessage(ref Message) ,
Control.ProcessKeyEventArgs(ref Message) , Control.ProcessKeyMessage(ref Message) ,
Control.RaiseDragEvent(object, DragEventArgs) , Control.RaisePaintEvent(object, PaintEventArgs) ,
Control.RecreateHandle() , Control.RectangleToClient(Rectangle) ,
Control.RectangleToScreen(Rectangle) , Control.ReflectMessage(nint, ref Message) ,
Control.Refresh() , Control.ResetMouseEventArgs() , Control.ResetText() , Control.ResumeLayout() ,
Control.ResumeLayout(bool) , Control.Scale(SizeF) , Control.Select() ,
Control.SelectNextControl(Control, bool, bool, bool, bool) , Control.SendToBack() ,
Control.SetBounds(int, int, int, int) , Control.SetBounds(int, int, int, int, BoundsSpecified) ,

Control.SizeFromClientSize(Size) , Control.SetStyle(ControlStyles, bool) , Control.SetTopLevel(bool) , Control.RtlTranslateAlignment(HorizontalAlignment) , Control.RtlTranslateAlignment(LeftRightAlignment) , Control.RtlTranslateAlignment(ContentAlignment) , Control.RtlTranslateHorizontal(HorizontalAlignment) , Control.RtlTranslateLeftRight(LeftRightAlignment) , Control.RtlTranslateContent(ContentAlignment) , Control.Show() , Control.SuspendLayout() , Control.Update() , Control.UpdateBounds() , Control.UpdateBounds(int, int, int, int) , Control.UpdateBounds(int, int, int, int, int, int) , Control.UpdateZOrder() , Control.UpdateStyles() , Control.OnImeModeChanged(EventArgs) , Control.AccessibilityObject , Control.AccessibleDefaultActionDescription , Control.AccessibleDescription , Control.AccessibleName , Control.AccessibleRole , Control.AllowDrop , Control.Anchor , Control.AutoScrollOffset , Control.LayoutEngine , Control.DataContext , Control.BackgroundImage , Control.BackgroundImageLayout , Control.Bottom , Control.Bounds , Control.CanFocus , Control.CanRaiseEvents , Control.CanSelect , Control.Capture , Control.CausesValidation , Control.CheckForIllegalCrossThreadCalls , Control.ClientRectangle , Control.CompanyName , Control.ContainsFocus , Control.ContextMenuStrip , Control.Controls , Control.Created , Control.Cursor , Control.DataBindings , Control.DefaultBackColor , Control.DefaultCursor , Control.DefaultFont , Control.DefaultForeColor , Control.DefaultMargin , Control.DefaultMaximumSize , Control.DefaultMinimumSize , Control.DefaultPadding , Control.DeviceDpi , Control.IsDisposed , Control.Disposing , Control.Dock , Control.DoubleBuffered , Control.Enabled , Control.Focused , Control.Font , Control.FontHeight , Control.ForeColor , Control.Handle , Control.HasChildren , Control.Height , Control.IsHandleCreated , Control.InvokeRequired , Control.IsAccessible , Control.IsAncestorSiteInDesignMode , Control.IsMirrored , Control.Left , Control.Margin , Control.ModifierKeys , Control.MouseButtons , Control.MousePosition , Control.Name , Control.Parent , Control.ProductName , Control.ProductVersion , Control.RecreatingHandle , Control.Region , Control.RenderRightToLeft , Control.ResizeRedraw , Control.Right , Control.RightToLeft , Control.ScaleChildren , Control.Site , Control.TabIndex , Control.TabStop , Control.Tag , Control.Top , Control.TopLevelControl , Control.ShowKeyboardCues , Control.ShowFocusCues , Control.UseWaitCursor , Control.Visible , Control.Width , Control.PreferredSize , Control.Padding , Control.ImeMode , Control.ImeModeBase , Control.PropagatingImeMode , Control.BackColorChanged , Control.BackgroundImageChanged , Control.BackgroundImageLayoutChanged , Control.BindingContextChanged , Control.CausesValidationChanged , Control.ClientSizeChanged , Control.ContextMenuStripChanged , Control.CursorChanged , Control.DockChanged , Control.EnabledChanged , Control.FontChanged , Control.ForeColorChanged , Control.LocationChanged , Control.MarginChanged , Control.RegionChanged , Control.RightToLeftChanged , Control.SizeChanged , Control.TabIndexChanged , Control.TabStopChanged , Control.TextChanged , Control.VisibleChanged , Control.Click ,

Control.ControlAdded , Control.ControlRemoved , Control.DataContextChanged , Control.DragDrop , Control.DragEnter , Control.DragOver , Control.DragLeave , Control.GiveFeedback , Control.HandleCreated , Control.HandleDestroyed , Control.HelpRequested , Control.Invalidated , Control.PaddingChanged , Control.Paint , Control.QueryContinueDrag , Control.QueryAccessibilityHelp , Control.DoubleClick , Control.Enter , Control.GotFocus , Control.KeyDown , Control.KeyPress , Control.KeyUp , Control.Layout , Control.Leave , Control.LostFocus , Control.MouseClick , Control.MouseDoubleClick , Control.MouseCaptureChanged , Control.MouseDown , Control.MouseEnter , Control.MouseLeave , Control.DpiChangedBeforeParent , Control.DpiChangedAfterParent , Control.MouseHover , Control.MouseMove , Control.MouseUp , Control.MouseWheel , Control.Move , Control.PreviewKeyDown , Control.Resize , Control.ChangeUICues , Control.StyleChanged , Control.SystemColorsChanged , Control.Validating , Control.Validated , Control.ParentChanged , Control.ImeModeChanged , Component.Dispose() , Component.GetService(Type) , Component.Container , Component.DesignMode , Component.Events , Component.Disposed , MarshalByRefObject.GetLifetimeService() , MarshalByRefObject.InitializeLifetimeService() , MarshalByRefObject.MemberwiseClone(bool) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Form1()

Initialises new instance of form class.

```
public Form1()
```

Methods

Dispose(bool)

Clean up any resources being used.

```
protected override void Dispose(bool disposing)
```

Parameters

`disposing` bool

true if managed resources should be disposed; otherwise, false.

UpdatePictureBox()

Initiates the picture box to redraw the bitmap by calling Invalidate().

```
public void UpdatePictureBox()
```

Class MoveTo

Namespace: ASE_Assignment

Assembly: ASE_Assignment.dll

Command which moves the pen to specified (x,y) coordinates.

```
public class MoveTo : CommandTwoParameters, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← MoveTo

Implements

ICommand

Inherited Members

CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed , CommandTwoParameters.CheckParameters(string[]) , CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Methods

Execute()

Calls BOOSE.Command.Execute(), then moves the pen to specified (x,y) coordinates.

```
public override void Execute()
```

See Also

CommandTwoParameters

Class PenColour

Namespace: ASE_Assignment

Assembly: ASE_Assignment.dll

Command which sets the pen colour to draw with using RGB values.

```
public class PenColour : CommandThreeParameters, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← CommandThreeParameters ← PenColour

Implements

ICommand

Inherited Members

CommandThreeParameters.param3 , CommandThreeParameters.param3unprocessed , CommandThreeParameters.CheckParameters(string[]) , CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed , CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Methods

Execute()

Calls BOOSE.Command.Execute(), then sets the pen colour to the specified RGB values.

```
public override void Execute()
```

See Also

CommandThreeParameters

Class Rect

Namespace: ASE_Assignment

Assembly: ASE_Assignment.dll

Command to draw a rectangle on the canvas.

```
public class Rect : CommandOneParameter, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← Rect

Implements

ICommand

Inherited Members

CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Rect()

Initialises new instance of Rect with no parameters.

```
public Rect()
```

Rect(int, int)

Initialises new instance of Rect using specified width and height.

```
public Rect(int width, int height)
```

Parameters

`width` int

Width of rectangle to be created

`height` int

Height of rectangle to be created

Methods

CheckParameters(string[])

Checks if the Rect command has been called with the correct amount of parameters.

```
public override void CheckParameters(string[] parameterList)
```

Parameters

`parameterList` string[]

String array to be checked to ensure the correct number of parameters has been passed.

Exceptions

CommandException

Thrown if the number of parameters is not 1.

Execute()

Calls BOOSE.Command.Execute(), then draws a rectangle with specified width and height.

```
public override void Execute()
```

See Also

CommandOneParameter

Namespace ASE_AssignmentTests

Classes

AppCanvasTests

Class AppCanvasTests

Namespace: ASE_AssignmentTests

Assembly: ASE_AssignmentTests.dll

```
[TestClass]
public class AppCanvasTests
```

Inheritance

object ← AppCanvasTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Setup()

Creates new canvas for testing purposes, uses specified width and height of 800px,800px.

```
[TestInitialize]
public void Setup()
```

TestDrawTo_SetsPenPosition()

Test to ensure DrawTo command correctly sets pen X and Y position after drawing.

```
[TestMethod]
public void TestDrawTo_SetsPenPosition()
```

TestMoveTo_SetsPenPosition()

Test to ensure MoveTo command correctly sets pen X and Y position.

```
[TestMethod]
public void TestMoveTo_SetsPenPosition()
```

TestMultilineProgram_NoRestrictions()

Test to ensure multiline programs can be run even if command parameters exceed BOOSE restrictions.

```
[TestMethod]
public void TestMultilineProgram_NoRestrictions()
```

TestMultilineProgram_WithRestrictions()

Test to ensure multiline programs can be run with command parameters that don't exceed BOOSE restrictions.

```
[TestMethod]
public void TestMultilineProgram_WithRestrictions()
```